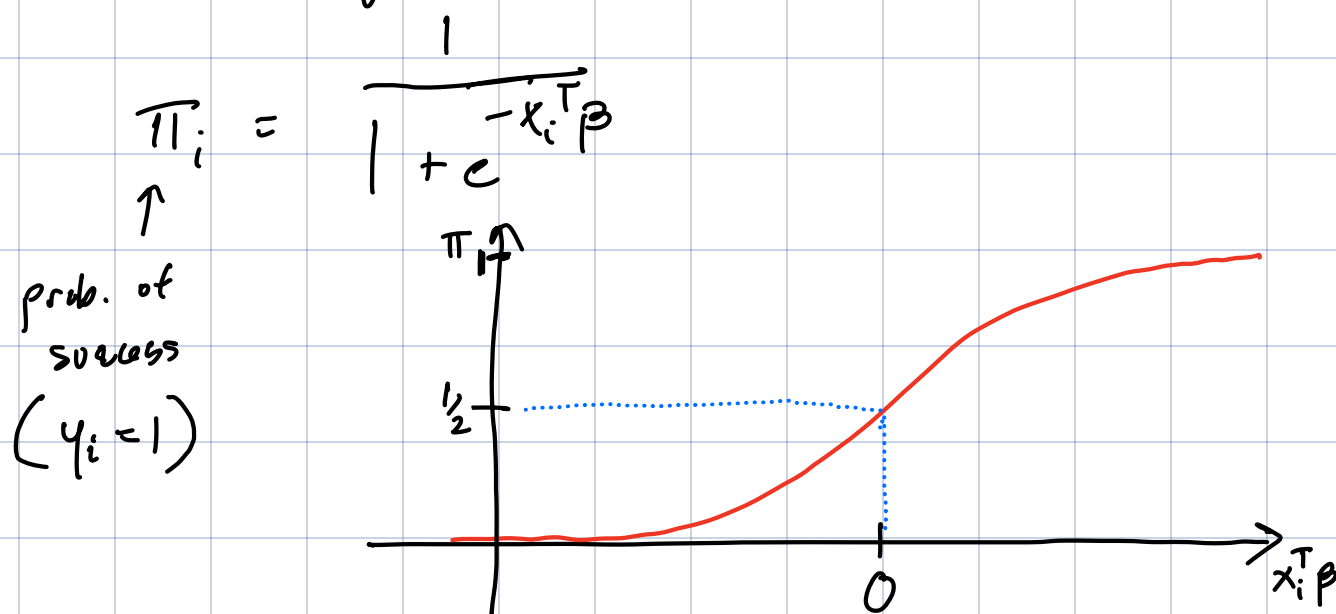


I Logistic Regression (cont.)



$$\pi_i = \frac{1}{2} \text{ if } x_i^T \beta = 0$$

$$\pi_i \rightarrow 0 \text{ as } x_i^T \beta \rightarrow -\infty$$

$$\pi_i \rightarrow 1 \text{ as } x_i^T \beta \rightarrow \infty$$

II Parameter Estimation

- We need MLE

$$P(Y_i = y_i) = \pi_i^{y_i} (1 - \pi_i)^{1 - y_i}$$

Value	Prob.
1	π_i
0	$1 - \pi_i$

$$\mathcal{L}(\beta) = \prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1 - y_i}$$

$$\ell(\beta) = \sum_{i=1}^n \ln \left(\pi_i^{y_i} (1 - \pi_i)^{1 - y_i} \right)$$

$$= \sum_{i=1}^n \left[y_i \ln \pi_i + (1-y_i) \ln (1-\pi_i) \right]$$

$$-\ell(\beta) = - \sum_{i=1}^n \dots$$

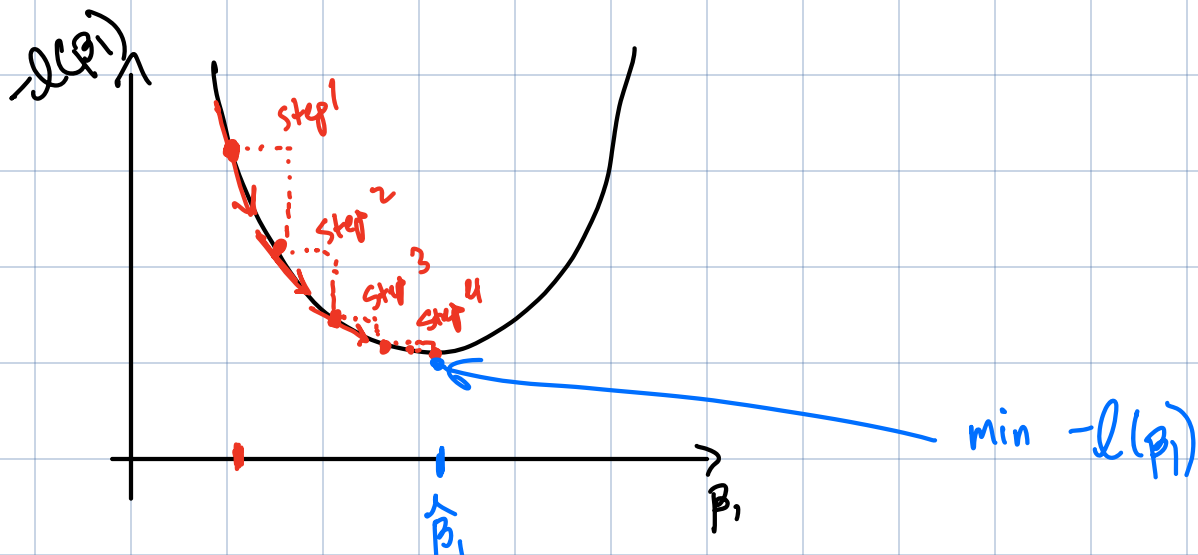
derivative of $-\ell(\beta) = -X^T (Y - \Pi)$

$$Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \end{pmatrix} \quad \Pi = \begin{pmatrix} \pi_1 \\ \pi_2 \\ \vdots \\ \pi_n \end{pmatrix}$$

No closed-form solution for β . Must use numerical optimization method.

Newton-Raphson method
Iteratively Reweighted Least Squares (IRLS)

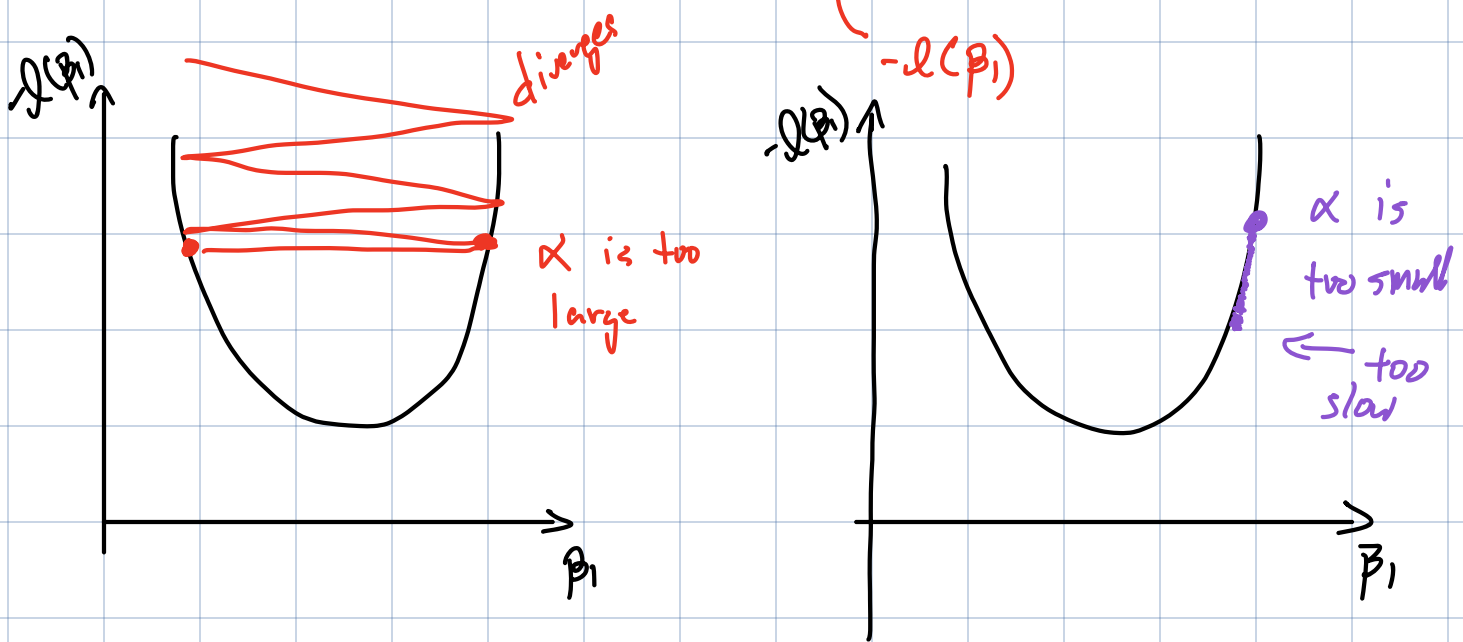
III Gradient Descent



The sign of the derivative tells us the direction. If $< 0 \Rightarrow$
 $> 0 \Leftarrow$
 $= 0$ STOP

The derivative has magnitude. Use this to help adjust β . For better control of our step size, let's introduce α , the learning rate.

$$\beta_1^{(t+1)} = \beta_1^{(t)} - \alpha \frac{d}{d\beta_1} \text{nl}(\beta_1)$$

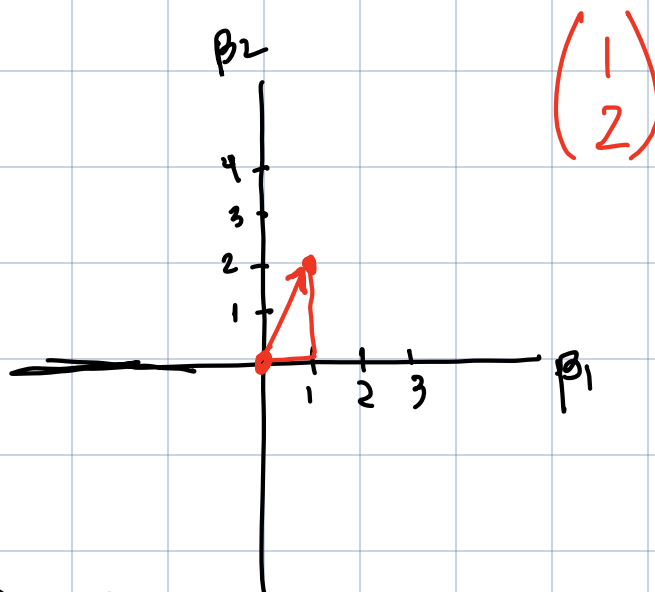


Gradient Descent 2D

Direction: we have more options, we need the steepest descent. The individual slopes are the partials $\frac{\partial}{\partial \beta_i} \text{nl}(\beta)$ combined into a vector is

the gradient $\nabla \text{nl}(\beta)$.

$$\begin{pmatrix} \frac{\partial}{\partial \beta_1} \text{nl}(\beta^{(t)}) \\ \frac{\partial}{\partial \beta_2} \text{nl}(\beta^{(t)}) \end{pmatrix}$$



Vanilla Gradient Descent

$$\beta^{(t+1)} = \beta^{(t)} - \alpha \nabla \text{nl}(\beta^{(t)})$$

$\nearrow -X^T (Y - \Pi)$